Red-Teaming Agentic AI: Exposing Risks and Validating Safeguards with Prediction Guard

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Prediction Guard

Introduction

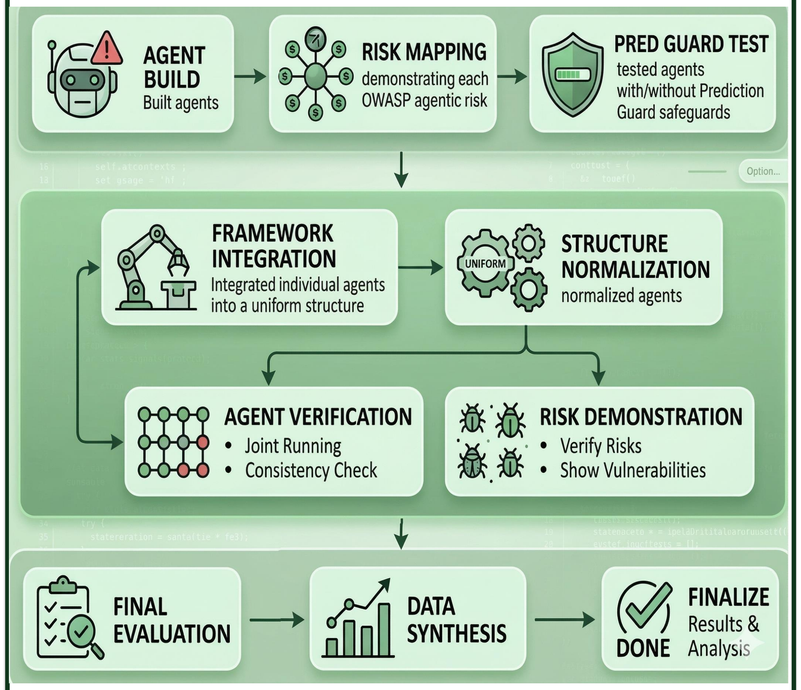
OVERVIEW

- Simulate security failures in agentic AI
- Implement 10 agents based on OWASP ASI risks
- Compare unsafe vs. Prediction Guard safeguarded agents

RELEVANCE

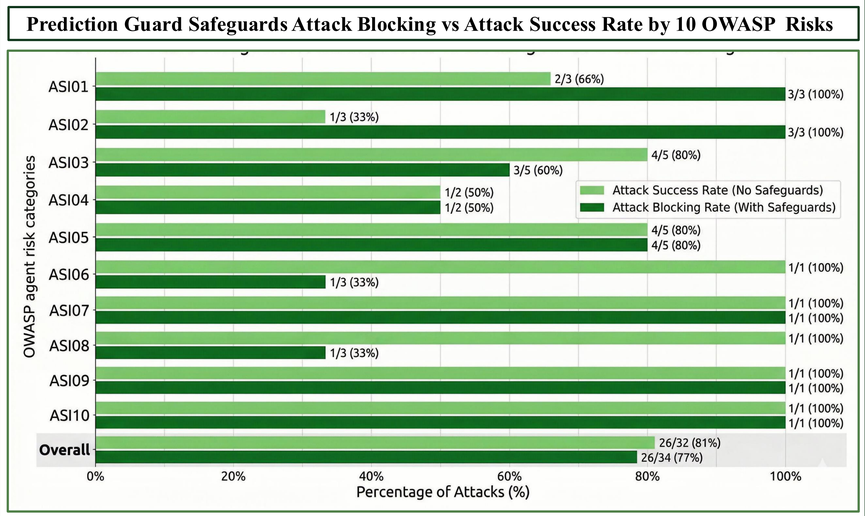
- Agentic AI is rapidly being deployed across industries
- Security vulnerabilities can lead to data exposure or unsafe actions
- Shows how Prediction Guard safeguards mitigate or reduce these risks

Methods



Results

Prediction Guard safeguards blocked 77% of attack cases across all 10 OWASP risk categories



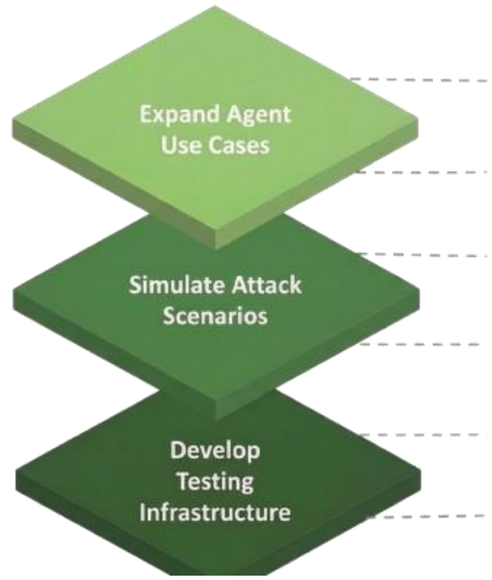
Model used to run the tests: **gpt-oss-120b**

- **81% Baseline Vulnerability:** Without Prediction Guard safeguards, 81% of attacks successfully bypassed the model's native security across all 10 OWASP categories.

- **100% Input Defense:** Neutralized all direct attacks for **Prompt Injection (ASI01)**, **PII Misuse (ASI02)**, **Inter-Agent Communication (ASI07)**, **Trust Exploitation (ASI09)**, **Rogue Agents (ASI10)**

- **Structural vs. Content Gaps:** Logic-heavy risks like **Memory Poisoning (ASI06)** and **Cascading Failures (ASI08)** saw a lower 33% block rate, highlighting the need for architectural safeguards beyond standard content filtering

Future Goals



- Increase test scenario depth for ASI03, ASI04, and ASI06
- Curate more structured datasets mirroring real examples.
- Conduct more runs with a secondary LLM to ensure cross-model consistency
- Use synthetic data to simulate attacks
- Create a detailed frontend for simplified user access.
- Develop logic-based layers to mitigate structural risks (ASI06/ASI08) beyond content filtering

Conclusion

INSIGHTS

- **Effective Input Defense:** Shifted the security posture from an 81% baseline failure rate to a 77% block rate, effectively neutralizing critical vulnerabilities.
- **Logic-Safeguard Gap:** Structural risks, specifically ASI06 and ASI08, showed a lower mitigation rate of 33% showing the need for architectural safeguards.

BUSINESS VALUE

Findings demonstrate that Prediction Guard reduces exploitable attack surface in production agentic systems with full audit trails and policy enforcement that regulated industries require.

Acknowledgements

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